1 Homogeneous Association Model I

Table 9.17 shows the result of cross-classifying a sample of people from the MBTI Step II National Sample, collected and compiled by CPP, Inc., on the four scales of the Myers-Briggs personality test: Extroversion/Introversion (E/I), Sensing/iNtuitive (S/N), Thinking/Feeling (T/F), and Judging/Perceiving (J/P). The 16 cells in this table correspond to the personality types. Fit the loglinear model of homogeneous association. Based on the fit, show that the estimated conditional association is strongest between the S/N and J/P scales and that there is not strong evidence of conditional association between the E/I and T/F scales or between the E/I and J/P scales.

Table 9.17 Data on Four Scales of Myers–Briggs Personality Test

Extroversion/Introversion	E				I			
Sensing/iNtuitive	S		N		S		N	
	Thinking/Feeling							
Judging/Perceiving	T	F	T	F	T	F	T	F
J	77	106	23	31	140	138	13	31
P	42	79	18	80	52	106	35	79

Source: Reproduced with special permission of CPP, Inc., Mountain View, CA 94043.

NTS: *One can tell whether someone is S or N by knowing if that person is J or P regardless of what the remaining letters are and vice versa. Knowing that one is E or I does not tell us if that same person is T or F or if that person is J or P.*

Table 1.1: MITB Dataset

EI	SN	TF	JP	count
E	N	T	J	23
E	N	T	P	18
E	N	F	J	31
E	N	F	P	80
I	N	T	J	13
I	N	T	P	35
I	N	F	J	31
I	N	F	P	79
E	S	T	J	77
E	S	T	P	42
E	S	F	J	106
E	S	F	P	79
I	S	T	J	140
I	S	T	P	52
I	S	F	J	138
I	S	F	P	106

$\lambda_{il}^{SN,JP}$ in

$$\log \mu_{ijkl} = \lambda + \lambda_i^{SN} + \lambda_j^{TF} + \lambda_k^{EI} + \lambda_l^{JP} + \lambda_{ij}^{SN,TF} + \lambda_{ik}^{SN,EI} + \lambda_{jk}^{TF,EI} + \lambda_{il}^{SN,JP} + \lambda_{jl}^{TF,JP} + \lambda_{kl}^{EI,JP}$$

represents the conditional association between SN and JP, holding all other variables (EI and TF) constant.

```
homogeneous_assoc_model <- loglm(count~(SN+TF+EI+JP)^2,
                                 data=prepared_df1,
                                 param=T,
                                 fit=T)

conditional_OR <- lapply(homogeneous_assoc_model$param, exp)
```

Table 1.2 shows that the estimated odds of being a Judger for Sensing types are 1.35715 times the odds for Intuitives (conditional odds ratio). The expected frequency of SJ / NP increases by -35.71476% compared to independence (multiplicative effect). Tables 1.3 and 1.4 show values close to 1.

Table 1.2: Conditional Odds Ratio: SN and JP

	J	P
N	0.73684	1.35715
S	1.35715	0.73684

Table 1.3: Conditional Odds Ratio: TF and EI

	E	I
F	1.04983	0.95253
T	0.95253	1.04983

Table 1.4: Conditional Odds Ratio: EI and JP

	J	P
E	1.00443	0.99559
I	0.99559	1.00443

Recall:

- Conditional odds ratio is simply the relationship between two variables while the other variables in the model are held constant
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  - OR = 1.45 implies that the first group has 45% greater odds of the outcome than the second group, or 1.45 times the odds of the second group.
  - OR = 0.81 implies that the first group has 19% lower odds of the outcome than the second group, or 0.81 times the odds of the second group.

2 Model selection

2.1 Anova

Refer to the previous exercise. Table 9.18 shows the fit of the model that assumes conditional independence between EI and TF and between EI and JP but has the other pairwise associations.

Table 9.18 Software Output (Based on SAS) for Fitting a Loglinear Model to Table 9.17

Criteria For Assessing Goodness Of Fit

Criterion	DF	Value
Deviance	7	12.3687
Pearson Chi-Square	7	12.1996

Analysis Of Parameter Estimates

Parameter			DF	Estimate	Standard Error	LR 95% Confidence Limits		Wald Chi-Square
EI*SN	e	n	1	0.3219	0.1360	0.0553	0.5886	5.60
SN*TF	n	f	1	0.4237	0.1520	0.1278	0.7242	7.77
SN*JP	n	j	1	−1.2202	0.1451	−1.5075	−0.9382	70.69
TF*JP	f	j	1	−0.5585	0.1350	−0.8242	−0.2948	17.12

Given SN, the variable EI is conditionally independent of TF and JP.

```
model_for_tab9p18 <- loglm(count~(SN+TF+EI+JP)^2 - EI:TF - EI:JP,
                           data=prepared_df1,
                           param=T,
                           fit=T)
```

```
anova(model_for_tab9p18, homogeneous_assoc_model)
```

```
LR tests for hierarchical log-linear models

Model 1:
count ~ (SN + TF + EI + JP)^2 - EI:TF - EI:JP
Model 2:
count ~ (SN + TF + EI + JP)^2

         Deviance df Delta(Dev) Delta(df) P(> Delta(Dev)
Model 1     12.369  7
Model 2     10.162  5      2.207         2        0.33171
Saturated    0.000  0     10.162         5        0.07078
```

It is enough to consider Model 1 for simplicity. This is because its P(> Delta(Dev) indicates Model 1 fits the data as good as Model 2 which is more complex due to the added parameters.

2.2 CI for conditional odds ratio

Show how to use the limits reported to construct a 95% profile likelihood confidence interval for the conditional odds ratio between the S/N and J/P scales. Interpret.

```
LB <- exp(-1.2202 - 1.96*0.1451)
UB <- exp(-1.2202 + 1.96*0.1451)
```

The CI is entirely below 1: (0.22211,0.39227). Hence, there is a negative association between SN & JP scales for any types of EI and TF.

2.3 DF

SAS (PROC GENMOD) reports maximized log-likelihood values of 3475.19 for the mutual independence model, 3538.05 for the homogeneous association model, and 3539.58 for the model containing all the three-factor interaction terms. Write the loglinear model for each case, and show that the numbers of model parameters are 5, 11, and 15, so residual df = 11,5, and 1.

Mutual independence model has one parameter for λ and $(2-1) = 1$ for each of λ_i^{SN}, λ_j^{TF}, λ_k^{EI}, and λ_l^{JP} which has two possible values, giving a total of 5 model parameters, resulting in 16 cells $-5 = 11$ as residual df.

$$\log \mu_{ijkl} = \lambda + \lambda_i^{SN} + \lambda_j^{TF} + \lambda_k^{EI} + \lambda_l^{JP}$$

Homogeneous association model has all the terms in mutual independence model. On top of that, each of the 6 additional addends increases the model df by $(2-1)(2-1) = 1$. That gives additional 6×1 more model parameters summing to 11, resulting in 16 cells $-11 = 5$ as residual df.

$$\log \mu_{ijkl} = \lambda + \lambda_i^{SN} + \lambda_j^{TF} + \lambda_k^{EI} + \lambda_l^{JP} + \lambda_{ij}^{SN,TF} + \lambda_{ik}^{SN,EI} + \lambda_{jk}^{TF,EI} + \lambda_{il}^{SN,JP} + \lambda_{jl}^{TF,JP} + \lambda_{kl}^{EI,JP}$$

The model for three-factor interaction considers all the terms in the homogeneous association model plus all possible combination of 3 variables which adds a total of 4 terms (SN-TF-JP, SN-TF-EI, SN-JP-EI, TF-EI-JP). Each of them contributes $(2-1)(2-1)(2-1) = 1$ additional model parameter which leads to $11 + 4 = 15$ total model parameters, resulting in 16 cells $-15 = 1$ as residual df.

2.4 AIC

According to AIC, which model in (c) seems best? Why?

Lower AIC is better when selecting models. Hence, we choose the homogeneous association model. See Table 2.1.

```
get_aic <- function(loglikelihood,
                    model_par_df){
  return(-2 * loglikelihood + 2 * model_par_df)
  }
```

```r
loglikelihoods <- c(3475.19, 3538.05, 3539.58)

`model parameter count` <- c(5, 11, 15)

model <- c("mutual independence",
           "homogeneous association",
           "3-factor interaction")

mods <- data.frame(model,
                   loglikelihoods,
                   `model parameter count`,
                   check.names = FALSE)

mods$AIC <- apply(mods, 1, function(x) {
  get_aic(
    as.numeric(x["loglikelihoods"]),
    as.numeric(x["model parameter count"])
    )
})
```

Table 2.1: AIC for Model Selection

model	loglikelihoods	model parameter count	AIC
mutual independence	3475.2	5	-6940.4
homogeneous association	3538.1	11	-7054.2
3-factor interaction	3539.6	15	-7049.2

3 Homogeneous Association Model II

Subjects in a GSS were asked their opinions about government spending on the environment (E), health (H), assistance to big cities (C), and law enforcement (L). The data are shown at the text website, with outcome categories 1 = too little, 2 = about right, 3 = too much. For the homogeneous association model, Table 9.19 shows some results, including the two-factor estimates for the EH association for coding by which estimates at category 3 of each variable equal 0.

Table 9.19 Software Output (Based on SAS) for Fitting Model For Exercise 9.5

Criteria For Assessing Goodness Of Fit

Criterion	DF	Value	Value/DF
Deviance	48	31.6695	0.6598
Pearson Chi-Square	48	26.5224	0.5526
Log Likelihood		1284.9404	

Parameter			DF	Estimate	Standard Error	Wald 95% Confidence Limits		Chi-Square
e*h	1	1	1	2.1425	0.5566	1.0515	3.2335	14.81
e*h	1	2	1	1.4221	0.6034	0.2394	2.6049	5.55
e*h	2	1	1	0.7294	0.5667	−0.3813	1.8402	1.66
e*h	2	2	1	0.3183	0.6211	−0.8991	1.5356	0.26

Table 9.20 Parameter Estimates for Model in Exercise 9.5

	Sum to Zero Constraints H			Zero for First Level H		
E	1	2	3	1	2	3
1	0.509	0.166	−0.676	0	0	0
2	−0.065	−0.099	0.163	0	0.309	1.413
3	−0.445	−0.068	0.513	0	0.720	2.142

3.1 Goodness of fit

Test the model goodness of fit, and interpret.

Both the deviance and Pearson chi-square are less than $\chi^2_{df=48} = 65.171$. Hence, the observed does not vary much from the expected - the model fits the data well.

$$\log \mu_{ijkl} = \lambda + \lambda_i^E + \lambda_j^H + \lambda_k^C + \lambda_l^L + \lambda_{ij}^{EH} + \lambda_{ik}^{EC} + \lambda_{jk}^{HC} + \lambda_{il}^{EL} + \lambda_{jl}^{HL} + \lambda_{kl}^{CL}$$

3.2 EH Conditional odds ratio

Report the estimated EH conditional odds ratio for the

(i) too much (3) and too little (1) categories,

$$\begin{aligned}
\log \theta_{13} &= \log \frac{\mu_{11}\mu_{33}}{\mu_{13}\mu_{31}} \\
&= \lambda_{11}^{EH} + \lambda_{33}^{EH} - \lambda_{13}^{EH} - \lambda_{31}^{EH} \\
&= 2.1425 + 0 - 0 - 0 \\
&= 2.1425
\end{aligned}$$

(ii) too much (3) and about right (2) categories, and

$$\begin{aligned}
\log\theta_{23} &= \log\frac{\mu_{22}\mu_{33}}{\mu_{23}\mu_{32}} \\
&= \lambda_{23}^{EH} + \lambda_{33}^{EH} - \lambda_{23}^{EH} - \lambda_{32}^{EH} \\
&= 0.3183 + 0 - 0 - 0 \\
&= 0.3183 \\
&\to \theta_{13} = 1.37479
\end{aligned}$$

(iii) about right (2) and too little (1) categories.

$$\begin{aligned}
\log\theta_{12} &= \log\frac{\mu_{11}\mu_{22}}{\mu_{12}\mu_{21}} \\
&= \lambda_{11}^{EH} + \lambda_{22}^{EH} - \lambda_{12}^{EH} - \lambda_{21}^{EH} \\
&= 2.1425 + 0.3183 - 1.4221 - 0.7294 \\
&= 0.3093 \\
&\to \theta_{13} = 1.36247
\end{aligned}$$

3.3 CI for odds ratio

Table 9.20 reports $\{\hat{\lambda}_{eh}^{EH}\}$ when parameters sum to zero within rows and within columns, and when parameters are zero in the first row and first column. Show how these yield the estimated EH conditional odds ratio for the too much and too little categories. Construct a confidence interval for that odds ratio. Interpret.

(From zero for first level)

$$\begin{aligned}
\log\theta_{13} &= \log\frac{\mu_{11}\mu_{33}}{\mu_{13}\mu_{31}} \\
&= \lambda_{11}^{EH} + \lambda_{33}^{EH} - \lambda_{13}^{EH} - \lambda_{31}^{EH} \\
&= 0.509 + 0.513 - (-0.676) - (-0.445) \\
&= 2.143 \\
&\to \theta_{13} = 8.52497
\end{aligned}$$

(From sum to zero constraint)

$$\begin{aligned}
\log\theta_{13} &= \log\frac{\mu_{11}\mu_{33}}{\mu_{13}\mu_{31}} \\
&= \lambda_{11}^{EH} + \lambda_{33}^{EH} - \lambda_{13}^{EH} - \lambda_{31}^{EH} \\
&= 0 + 2.142 - 0 - 0 \\
&= 2.142 \\
&\to \theta_{13} = 8.51645
\end{aligned}$$

```
LB <- exp(2.1425 - 1.96*0.5566)
UB <- exp(2.1425 + 1.96*0.5566)
```

The CI is entirely above 1: (2.86212,25.36667). There is a positive association between opinions about government spending for Education and Health.

4 Proving I

For an $I \times J$ table, let $\eta_{ij} = \log \mu_{ij}$, and let a dot subscript alone denote the mean for that index (i.e., $\eta_{i.} = \sum_j \eta_{ij}/J$). Let $\lambda = \eta_{..}$, $\lambda_i^X = \eta_{i.} - \eta_{..}$, $\lambda_i^Y = \eta_{.j} - \eta_{..}$ and $\lambda_{ij}^{XY} = \eta_{ij} - \eta_{i.} - \eta_{.j} + \eta_{..}$

4.1 I.1

Show that $\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_{ij}^{XY}$. Hence, any set of positive $\{\mu_{ij}\}$ satisfies the saturated model

$$\begin{aligned}
\log \mu_{ij} &= \lambda + \lambda_i^X + \lambda_j^Y + \lambda_{ij}^{XY} \\
&= \eta_{..} + \eta_{i.} - \eta_{..} + \eta_{.j} - \eta_{..} + \eta_{ij} - \eta_{i.} - \eta_{.j} + \eta_{..} \\
&= \eta_{ij}
\end{aligned}$$

4.2 I.2

Show that $\sum_i \lambda_i^X = \sum_j \lambda_j^Y = \sum_i \lambda_{ij}^{XY} = \sum_j \lambda_{ij}^{XY} = 0$

$$\begin{aligned}
\sum_i \lambda_i^X &= \sum_i \eta_{i.} - \eta_{..} \\
&= \sum_i \eta_{i.} - \sum_i \eta_{..} \\
&= \sum_i \eta_{i.} - I\eta_{..} \\
&= \sum_i \eta_{i.} - I\frac{\sum_i \eta_{i.}}{I} \\
&= \sum_i \eta_{i.} - \sum_i \eta_{i.} \\
&= 0
\end{aligned}$$

$$
\begin{aligned}
\sum_j \lambda_j^X &= \sum_j \eta_{.j} - \eta_{..} \\
&= \sum_j \eta_{.j} - \sum_j \eta_{..} \\
&= \sum_j \eta_{.j} - J\eta_{..} \\
&= \sum_j \eta_{.j} - J\frac{\sum_j \eta_{.j}}{J} \\
&= \sum_j \eta_{.j} - \sum_j \eta_{.j} \\
&= 0
\end{aligned}
$$

$$
\begin{aligned}
\sum_j \lambda_{ij}^{XY} &= \sum_j (\eta_{ij} - \eta_{i.} - \eta_{.j} + \eta_{..}) \\
&= \sum_j \eta_{ij} - \sum_j \eta_{i.} - \sum_j \eta_{.j} + \sum_j \eta_{..} \\
&= J\eta_{i.} - J\eta_{i.} - J\eta_{..} + J\eta_{..} \\
&= 0
\end{aligned}
$$

$$
\begin{aligned}
\sum_i \lambda_{ij}^{XY} &= \sum_i (\eta_{ij} - \eta_{i.} - \eta_{.j} + \eta_{..}) \\
&= \sum_i \eta_{ij} - \sum_i \eta_{i.} - \sum_i \eta_{.j} + \sum_i \eta_{..} \\
&= I\eta_{.j} - I\eta_{..} - I\eta_{.j} + I\eta_{..} \\
&= 0
\end{aligned}
$$

4.3 I.3

For 2×2 tables, show that $\log \theta = 4\lambda_{11}^{XY}$

2×2 tables showing estimated parameters are symmetric: $\lambda_{11}^{XY} = \lambda_{21}^{XY}$. Values in the off-diagonals have the same magnitude as λ_{11}^{XY} but are negative. These are consequences of the sum to zero constraint.

$$
\begin{aligned}
\log \theta_{12} &= \log \frac{\mu_{11}\mu_{22}}{\mu_{12}\mu_{21}} \\
&= \lambda_{11}^{XY} + \lambda_{22}^{XY} - \lambda_{12}^{XY} - \lambda_{21}^{XY} \\
&= \lambda_{11}^{XY} + \lambda_{11}^{XY} - (-\lambda_{11}^{XY}) - (-\lambda_{11}^{XY}) \qquad \text{sum to zero constraint} \\
&= \lambda_{11}^{XY} + \lambda_{11}^{XY} + \lambda_{11}^{XY} + \lambda_{11}^{XY} \\
&= 4\lambda_{11}^{XY}
\end{aligned}
$$

4.4 I.4

For $2 \times J$ tables, show that $\lambda_{11}^{XY} = \left(\Sigma_j \log \alpha_j\right)/2J$, where $\alpha_j = \frac{\mu_{11}\mu_{2j}}{\mu_{21}\mu_{1j}},\ j = 2, \ldots, J$

$$
\begin{aligned}
\frac{\left(\sum_j \log \frac{\mu_{11}\mu_{2j}}{\mu_{21}\mu_{1j}}\right)}{2J} &= \frac{\left(\sum_j \left[\lambda_{11}^{XY} + \lambda_{2j}^{XY} - \lambda_{1j}^{XY} - \lambda_{21}^{XY}\right]\right)}{2J} \\
&= \frac{\left(\sum_j \lambda_{11}^{XY} + \sum_j \lambda_{2j}^{XY} - \sum_j \lambda_{1j}^{XY} - \sum_j \lambda_{21}^{XY}\right)}{2J} \\
&= \frac{\left(J\lambda_{11}^{XY} + \sum_j \lambda_{2j}^{XY} - \sum_j \lambda_{1j}^{XY} - J\lambda_{21}^{XY}\right)}{2J} \\
&= \frac{\left(J\lambda_{11}^{XY} + 0 - 0 - J\lambda_{21}^{XY}\right)}{2J} \qquad \text{sum to zero constraint} \\
&= \frac{J\left(\lambda_{11}^{XY} - \lambda_{21}^{XY}\right)}{2J} \\
&= \frac{\left(\lambda_{11}^{XY} - \lambda_{21}^{XY}\right)}{2}
\end{aligned}
$$

$$\lambda_{11}^{XY} + \lambda_{21}^{XY} = 0 \rightarrow \lambda_{21}^{XY} = -\lambda_{11}^{XY} \qquad \text{sum to zero constraint}$$

Continuing,

$$
\begin{aligned}
&= \frac{\left(\lambda_{11}^{XY} - (-\lambda_{11}^{XY})\right)}{2} \\
&= \frac{2\left(\lambda_{11}^{XY}\right)}{2} \\
&= \lambda_{11}^{XY}
\end{aligned}
$$

5 Proving II

For three categorical variables X, Y, and Z:

5.1 Joint to conditional independence

When Y is jointly independent of X and Z, show that X and Y are conditionally independent, given Z.

$$P(x, y, z) = P(y)P(x, z) \rightarrow P(x, y \mid z) = P(x \mid z)P(y \mid z)$$

$$
\begin{aligned}
P(y,z) &= \sum_x P(x,y,z) \\
&= \sum_x P(y)P(x,z) \qquad \text{assumption} \\
&= P(y)\sum_x P(x,z) \qquad \text{y is constant relative to x} \\
&= P(y)P(z) \qquad \text{summing across x rows in column z} \\
\rightarrow P(y) &= \frac{P(y,z)}{P(z)} \\
&= P(y \mid z) \qquad \text{def of conditional prob} \\
P(x,y \mid z) &= \frac{P(x,y,z)}{P(z)} \qquad \text{def of conditional prob} \\
&= \frac{P(y)P(x,z)}{P(z)} \qquad \text{assumption} \\
&= P(y)P(x \mid z) \qquad \text{def of conditional prob} \\
&= P(y \mid z)P(x \mid z)
\end{aligned}
$$

5.2 Mutual to marginal & conditional independence

Prove that mutual independence of X, Y, and Z implies that X and Y are both marginally and conditionally independent.

$$P(x,y,z) = P(y)P(x)P(z) \rightarrow P(x,y) = P(x)P(y) \quad \text{(marginal independence)}$$

$$
\begin{aligned}
P(x,y) &= \sum_z P(x,y,z) \\
&= \sum_z P(y)P(x)P(z) \qquad \text{assumption} \\
&= P(y)P(x)\sum_z P(z) \qquad \text{y, x is constant relative to z} \\
&= P(y)P(x)\cdot 1 \qquad \text{sure event} \\
&= P(x)P(y)
\end{aligned}
$$

$$P(x,y,z) = P(y)P(x)P(z) \rightarrow P(x,y \mid z) = P(x \mid z)P(y \mid z) \quad \text{(conditional independence)}$$

$$
\begin{aligned}
P(x, y \mid z) &= \frac{P(x, y, z)}{P(z)} && \text{def of conditional prob} \\
&= \frac{P(y)P(x)P(z)}{P(z)} && \text{by assumption (mutual ind)} \\
&= P(y)P(x) \\
&= P(y \mid z)P(x \mid z) && \text{by assumption (mutual ind)}
\end{aligned}
$$

5.3 Pairwise and mutual independence

When X is independent of Y and Y is independent of Z, explain why X is not necessarily independent of Z.

This means that pairwise independence does not imply mutual independence even if mutual independence implies pairwise independence. It is enough to provide a counterexample: Math (X) and English (Y) proficiency levels may be independent, and English (Y) and Science (Z) proficiency levels may be independent, but Science (Z) often involve math skills (X), so Math and Science proficiency levels are not necessarily independent.